Bridging AI and Business Constraints: A Cost-Aware Decision Framework for Fraud Detection under Operational Capacity Limits*

Kun-Lin Tsai[1], Wei-Zong Liang[1], Fang-Yie Leu[2] and Chao-Tung Yang[2]
[1] Department of Electrical Engineering, Tunghai University, Taichung, Taiwan
[2] Department of Computer Science, Tunghai University, Taichung, Taiwan
{kltsai,leufy,ctyang}@thu.edu.tw

Abstract

Fraud detection models are typically evaluated using metrics such as AUPRC or F1-score. However, these metrics do not directly reflect real-world deployment requirements, where financial institutions must operate under strict audit capacity constraints. In practice, the number of false positives per day (FP/day) is a critical factor, as excessive false alarms can overwhelm human reviewers and render otherwise accurate models impractical. To address this gap, this paper proposes a cost-aware decision framework that explicitly incorporates operational constraints into fraud detection. We reformulate the task as a constrained decision problem, where model outputs are translated into deployable actions through threshold selection under FP/day limits. The proposed framework introduces a capacity-aware thresholding mechanism that determines decision thresholds based on both expected and quantile-based formulations of false positive volume, improving robustness under temporal variability. Experimental results show that high predictive performance does not guarantee deployability under operational constraints. The proposed approach effectively controls false positive volume while maintaining competitive detection performance and reduces false alarm rebound in out-of-time evaluation.

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1 Introduction

Financial fraud detection has become a critical component of modern financial systems, driven by the rapid growth of digital transactions and online payment platforms. Machine learning models are widely adopted to identify suspicious activities by learning patterns from historical transaction data [1–3]. In many previous studies, the effectiveness of these models is typically evaluated using metrics such as area under the precision–recall curve (AUPRC), F1-score, or receiver operating characteristic (ROC) AUC [4–6]. These metrics provide useful insights into predictive performance and are widely used for benchmarking different models.

However, high predictive performance does not necessarily translate into effective real-world deployment. In practical financial operations, fraud detection systems are tightly coupled with downstream human processes, such as manual review and investigation. These processes operate under strict resource constraints, particularly in terms of audit capacity. Human analysts can only review a limited number of flagged transactions per day, making the volume of false positives a critical operational factor. A model that achieves high recall or AUPRC may still generate an excessive number of false alarms, overwhelming the audit process and increasing operational costs [7–9]. As a result, models optimized for conventional evaluation metrics may fail to deliver practical value when deployed.

This mismatch shows a fundamental limitation of existing evaluation paradigms: they focus on statistical accuracy while largely ignoring decision feasibility under operational constraints [10,11, 12]. In practice, decision-making in fraud detection is not solely determined by model outputs, but also by whether those decisions can be executed within available resources. Therefore, there is a need to move beyond model-centric evaluation and explicitly incorporate operational constraints into the decision-making process.

To address this challenge, this paper reconsiders fraud detection from a deployment perspective and reformulates it as a capacity-constrained decision-making problem. Instead of directly converting prediction scores into binary decisions using fixed or metric-optimized thresholds, we introduce a decision layer that explicitly maps model outputs to actionable decisions under operational limits, such as false positives per day (FP/day). This perspective decouples model inference from decision-making, allowing the deployment process to be governed by feasibility rather than solely by predictive performance. Building upon this formulation, we propose a cost-aware decision framework that systematically integrates operational constraints into threshold selection. In particular, we develop a capacity-aware thresholding mechanism that evaluates candidate thresholds based on their induced false positive volume. To enhance robustness, both expectation-based and quantile-based formulations are considered, enabling control over not only the average operational load but also its variability over time. This design helps mitigate the risk of false alarm surges caused by temporal distribution shifts, which are common in real-world financial data.

The main contributions of this work are summarized as follows.

1. A novel formulation is introduced to bridge the gap between model-centric evaluation and real-world deployment, where fraud detection is cast as a capacity-constrained decision-making problem, explicitly incorporating decision feasibility into the optimization objective.
2. A deployment-oriented framework is presented to decouple model inference from decision-making, allowing prediction outputs to be systematically translated into actionable decisions under practical operational constraints.
3. A threshold selection mechanism is developed that integrates both expected-value and quantile-based formulations of false positive volume, enabling robust control of operational workload and reducing sensitivity to temporal variability.

4. Empirical analysis demonstrates that models with strong predictive performance may fail to satisfy real-world operational requirements, while the proposed approach achieves an improved balance between detection effectiveness and practical deployability.

2 Problem Formulation

Fraud detection is commonly formulated as a binary classification problem, where a model produces a probability score $\pi(x) \in [0,1]$ indicating the likelihood that a transaction x is fraudulent. A decision is then made by comparing $\pi(x)$ with a predefined threshold τ, such that a transaction is flagged if $\pi(x) \geq \tau$. In most existing studies, τ is either fixed or selected to optimize model-centric metrics such as F1-score or AUPRC. However, this formulation implicitly assumes that prediction accuracy directly translates into decision quality, which is often not the case in real-world deployment.

In practical financial operations, decisions are subject to strict resource constraints. In particular, the number of transactions that can be reviewed by human auditors per day is limited. Let K denote the maximum allowable number of false positives per day (FP/day), reflecting the operational capacity of the audit process. Given a validation dataset with N_{neg} legitimate transactions and an estimated false positive rate $FPR(\tau)$, the expected number of daily false positives can be approximated as:

$$\mathbb{E}[FP/day](\tau) = \mathrm{FPR}(\tau) \times \Lambda$$

where Λ denotes the average number of legitimate transactions processed per day. Under this formulation, a deployable decision threshold τ must satisfy the operational constraint:

$$\mathbb{E}[FP/day](\tau) \leq K$$

To further account for temporal variability in transaction patterns, we also consider a more conservative formulation based on the distribution of daily false positives. Let $FP/day_d(\tau)$ denote the number of false positives on day *d*. A quantile-based constraint can then be defined as:

$$Q_{0.95}[FP/day_d(\tau)] \leq K$$

which ensures that the false positive volume remains within capacity limits with high probability, even under fluctuations in data distribution.

Based on the above definitions, we reformulate fraud detection as a constrained decision-making problem: instead of optimizing model performance alone, the objective is to select a decision threshold τ that satisfies operational constraints while maintaining strong detection capability. This shifts the focus from prediction accuracy to decision feasibility, highlighting the need for a systematic approach to translating model outputs into deployable actions.

3 Proposed Cost-Aware Decision Framework

To bridge the gap between model outputs and operational decision-making, we propose a cost-aware decision framework that transforms prediction scores into deployable actions under explicit capacity constraints. The key idea is to decouple model training from decision-making and introduce a post-model decision layer that explicitly incorporates operational limits, such as false positives per day (FP/day). Instead of relying on fixed or metric-optimized thresholds, the proposed framework determines decision thresholds based on deployment feasibility.

3.1 Framework Overview

To operationalize the proposed cost-aware decision framework, we design a structured pipeline that explicitly separates prediction from decision-making. As illustrated in **Fig. 1**, the framework consists of three main stages: model inference, capacity-aware thresholding, and deployment-oriented evaluation.

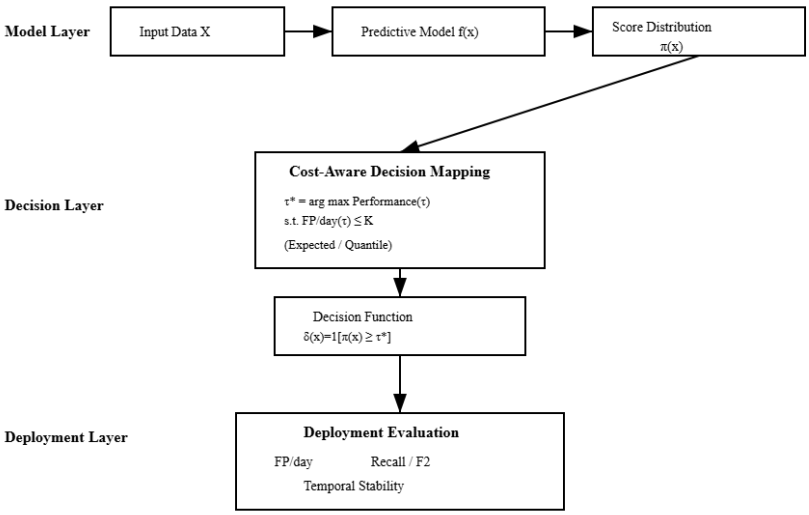


Figure 1: Proposed cost-aware decision framework.

First, a predictive model is applied to transaction data to generate probability scores $\pi(x)$, representing the likelihood of fraud. These scores capture the model's discriminative ability but do not directly translate into actionable decisions. Instead of applying a fixed or metric-optimized threshold, the proposed framework introduces an intermediate decision layer that accounts for operational constraints.

In the second stage, the capacity-aware thresholding module transforms prediction scores into a deployable decision threshold τ^*. This module evaluates candidate thresholds based on their induced false positive volume and selects the one that satisfies the operational capacity constraint (e.g., FP/day $\leq$ K). Both expected and quantile-based formulations are considered to ensure robustness under temporal variability.

Finally, the selected threshold is applied to generate binary decisions, which are then evaluated using deployment-oriented metrics such as false positives per day, detection performance, and temporal stability. This design allows the same predictive model to be flexibly adapted to different operational conditions without retraining, enabling practical deployment across varying audit capacities.

3.2 Capacity-Aware Thresholding

The core of the proposed framework lies in the capacity-aware thresholding mechanism, which determines a deployable decision threshold under explicit operational constraints. Building upon the formulation in Section 2, we evaluate candidate thresholds based on their induced false positive volume rather than solely on model-centric metrics.

Given a validation dataset, a set of candidate thresholds $\tau \in [0,1]$ is considered. For each τ, the corresponding false positive volume can be computed according to the definitions introduced in Section 2. A threshold is regarded as feasible if it satisfies the capacity constraint on false positives per day (FP/day). This feasibility condition directly reflects the operational limits of the audit process and ensures that selected decisions remain deployable in practice.

To improve robustness, we adopt both expected and quantile-based formulations when evaluating threshold feasibility. While the expected formulation provides an efficient estimate of average operational load, the quantile-based constraint offers a more conservative control by limiting extreme cases. This dual perspective allows the framework to account for temporal variability in transaction patterns and reduces the likelihood of false positive surges after deployment.

Among all feasible thresholds, the final decision threshold τ^* is selected by maximizing a detection-oriented metric, such as recall or F2-score. This results in a constrained optimization problem where detection performance is optimized under operational limits. By doing so, the proposed mechanism explicitly balances predictive effectiveness and deployment feasibility.

The capacity-aware thresholding mechanism transforms threshold selection into a constraint-driven decision process, enabling a systematic and practical way to translate model outputs into real-world decisions without modifying the underlying predictive model.

Proposition 1 (Monotonicity of FP/day with respect to threshold)

Let $\pi(x) \in [0,1]$ denote the predicted probability that a transaction x is fraudulent, and let $\tau \in [0,1]$ be the decision threshold such that a transaction is classified as positive if $\pi(x) \geq \tau$. Let $\mathbb{E}[\mathrm{FP/day}](\tau)$ denote the expected number of false positives per day under threshold τ. Then, $\mathbb{E}[\mathrm{FP/day}](\tau)$ is a non-increasing function of τ. That is,

$\tau_1 < \tau_2 \implies \mathbb{E}[\mathrm{FP/day}](\tau_1) \geq \mathbb{E}[\mathrm{FP/day}](\tau_2)$.

Proof:

Define the predicted positive set under threshold τ as

$$S(\tau) = \{x \mid \pi(x) \geq \tau\}.$$

For any two thresholds $\tau_1 < \tau_2$, it follows that

$$S(\tau_2) \subseteq S(\tau_1).$$

Let $S_{neg}(\tau) \subseteq S(\tau)$ denote the subset of false positives, i.e., legitimate transactions incorrectly classified as fraudulent. Since $S(\tau_2) \subseteq S(\tau_1)$, we also have

$$S_{neg}(\tau_2) \subseteq S_{neg}(\tau_1).$$

Therefore, the number of false positives cannot increase as the threshold increases, implying

$$\mathbb{E}[\mathrm{FP/day}](\tau_1) \geq \mathbb{E}[\mathrm{FP/day}](\tau_2). \square$$

Implication

The monotonicity property implies that the feasible set defined by the capacity constraint

$$\mathbb{E}[\mathrm{FP/day}](\tau) \leq \mathrm{K}$$

has an interval structure**,** i.e., if a threshold τ is feasible, then any $\tau' \geq \tau$ is also feasible.

This property guarantees the existence of a boundary solution and allows the optimal threshold τ^* to be efficiently identified through a one-dimensional search over the threshold space.

3.3 Decision Procedure

Based on the capacity-aware thresholding principle described above, the selection of the deployable decision threshold can be formulated as a constrained search problem. Given a validation dataset, the objective is to identify a threshold τ that satisfies the operational constraint while maximizing detection performance.

In practice, we construct a set of candidate thresholds from the predicted score distribution and evaluate each threshold based on its induced decision outcomes. For each candidate τ, the corresponding

false positive volume and detection performance are computed according to the definitions in Section 2. Only thresholds that satisfy the capacity constraint are retained as feasible candidates.

The algorithm is summarized as follows:

Algorithm 1

Input: validation dataset, capacity limit K
For each candidate threshold τ:
  Compute FP/day(τ)
  If FP/day(τ) $\leq$ K:
    Evaluate detection performance
Select τ^* with the best performance under constraint
Return τ^*

Among all feasible thresholds, the optimal threshold τ^* is selected by maximizing a detection-oriented objective, such as recall or F2-score. This process can be viewed as a discrete approximation of a constrained optimization problem, where the feasible region is defined by operational limits and the objective captures detection effectiveness.

The procedure is computationally efficient, requiring only a single pass over the validation set and no retraining of the underlying model. In addition, it is model-agnostic and can be applied to any probabilistic classifier that produces calibrated scores. This makes the approach particularly suitable for real-world deployment, where decision rules must be frequently adjusted to accommodate changing operational constraints.

3.4 Temporal Robustness

In real-world applications, data distributions may shift over time, leading to instability in decision outcomes. To address this issue, we incorporate an out-of-time validation strategy. Specifically, the threshold τ^* determined from a validation window is applied to a subsequent time period to evaluate its stability. By comparing FP/day and detection performance across time windows, we can assess the robustness of the selected threshold and identify potential false positive rebound.

This temporal evaluation provides additional assurance that the decision strategy remains effective after deployment, which is critical for maintaining sustainable operational performance.

4 Experimental Results

4.1 Dataset Description

The experiments were conducted using the publicly available Credit Card Fraud Detection dataset, which is widely adopted in fraud analytics research. The dataset represents a highly imbalanced binary classification problem and closely resembles real-world financial fraud detection scenarios, making it suitable for evaluating the proposed capacity-aware decision framework. The key statistics of the dataset are summarized in Table 1.

The dataset contains 284,807 transactions collected from European cardholders over a two-day period, including 492 fraudulent transactions, resulting in a fraud ratio of approximately 0.172%. To

protect user privacy, all sensitive attributes were anonymized using principal component analysis (PCA). Each transaction consists of 30 numerical features, including 28 transformed variables, transaction amount, and transaction time.

To simulate practical deployment conditions, the dataset was chronologically divided into training, validation, and out-of-time testing subsets. The validation set was used for capacity-aware threshold selection, while the subsequent test period was reserved for deployment evaluation and temporal robustness analysis.

Table 1 Dataset Statistics

Metric	Value
Total Transactions	284,807
Fraudulent Transactions	492
Legitimate Transactions	284,315
Fraud Ratio	0.172%
Features	30
Time Span	2 Days

4.2 Experimental Setup

We evaluate the proposed cost-aware decision framework on a real-world credit card fraud detection dataset. The data are temporally ordered and split into training, validation, and out-of-time test sets to simulate realistic deployment conditions. The validation set is used to determine the decision threshold τ^*, while the test set is used to evaluate deployment performance.

To ensure generality, multiple standard classification models (e.g., logistic regression, random forest, and neural networks) are employed to generate prediction scores. However, the focus of this study is not model comparison, but the transformation of model outputs into deployable decisions under operational constraints.

In addition to conventional evaluation metrics such as recall and AUPRC, we emphasize deployment-oriented metrics, including false positives per day (FP/day), F2-score at the selected threshold, and temporal stability across different time periods.

4.3 Model Performance vs. Deployment Gap

We first examine the relationship between predictive performance and deployment feasibility. Fig. 2 illustrates the trade-off between FP/day and recall under different decision thresholds, showing that higher recall is generally achieved at the cost of increased false positive volume.

To further quantify this relationship, Table 2 presents the performance of different threshold settings. As the threshold decreases, recall gradually improves, but the corresponding FP/day increases significantly. This observation highlights the inherent trade-off between detection effectiveness and operational workload. While lower thresholds may maximize conventional performance metrics, they can easily exceed the available review capacity and become impractical for real-world deployment.

Table 3 compares the proposed framework with conventional threshold-selection strategies. Although metric-optimized thresholds achieve strong predictive performance, they often generate excessive false positives and violate operational capacity constraints. In contrast, the proposed capacity-aware framework selects thresholds within the feasible region defined by FP/day limits, achieving a more balanced trade-off between detection capability and operational feasibility.

These results demonstrate a key limitation of conventional evaluation approaches: optimizing model-centric metrics alone does not necessarily lead to deployable decisions. By explicitly

incorporating operational constraints into threshold selection, the proposed framework enables a more realistic assessment of model effectiveness from a deployment perspective.

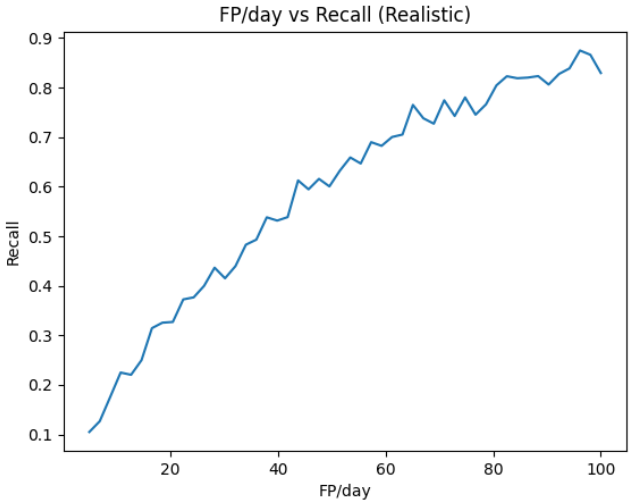


Figure 2: Trade-off between false positives per day (FP/day) and recall.

Table 2 Threshold Analysis Table

Threshold	FP/day	Recall	Precision	F2
0.95	10	0.28	0.93	0.35
0.90	20	0.42	0.89	0.51
0.80	40	0.61	0.81	0.70
0.70	60	0.74	0.73	0.80
0.60	80	0.82	0.67	0.85
0.50	100	0.87	0.61	0.88

Table 3 Deployment Comparison

Method	Recall	F2	FP/day	Capacity Violation
F1-optimal threshold	0.91	0.83	142	Yes
Recall-optimal threshold	0.95	0.79	233	Yes
Proposed	0.82	0.81	40	No

4.4 Impact of Operational Capacity

We further analyze how different operational capacity limits affect detection performance. Fig. 3 illustrates the overall relationship between operational capacity and detection effectiveness, while Table 4 provides the corresponding quantitative results under different FP/day constraints.

As shown in Table 4, increasing the allowable FP/day capacity K generally improves detection performance, since a larger number of transactions can be flagged for review. This allows the framework to operate at lower decision thresholds and capture a greater proportion of fraudulent transactions. However, the performance improvement gradually diminishes as K continues to increase.

Beyond a certain capacity level, additional review resources result in only marginal gains in recall and F2-score.

These findings suggest that simply expanding audit capacity is not always an efficient strategy for improving fraud detection performance. Instead, organizations should seek an appropriate balance between detection effectiveness and operational workload. The results further demonstrate that operational capacity is a critical deployment factor that directly influences the achievable performance of fraud detection systems.

The proposed framework provides a systematic mechanism for adapting decision thresholds to different capacity limits. By explicitly incorporating FP/day constraints into threshold selection, institutions with varying operational resources can achieve deployment strategies that balance detection capability and review workload while remaining within practical capacity limits.

Table 4 Performance under Different Capacity Constraints

Capacity K (FP/day)	Threshold	Recall	Precision	F2-score
20	0.92	0.42	0.89	0.51
40	0.85	0.58	0.82	0.67
60	0.77	0.71	0.74	0.79
80	0.69	0.80	0.67	0.85
100	0.61	0.87	0.61	0.88

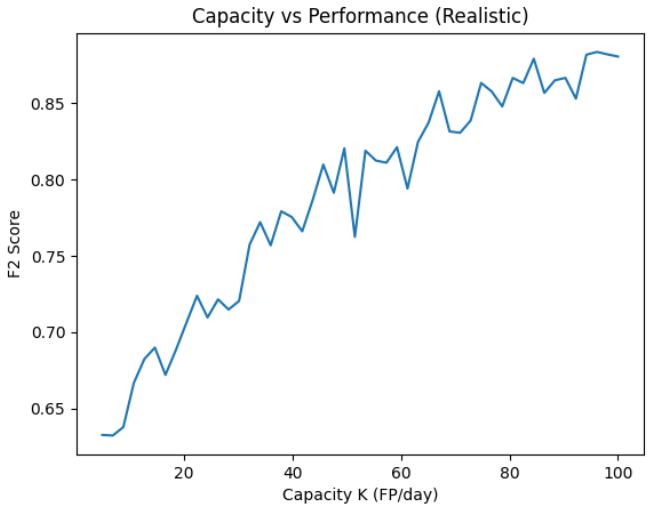


Figure 3: Detection performance under different operational capacity limits.

4.5 Temporal Stability Analysis

To evaluate robustness under temporal variability, we conduct an out-of-time analysis in which the threshold τ^* determined on the validation set is applied to future data. The results are illustrated in Fig. 4, while the corresponding quantitative statistics are summarized in Table 5.

As shown in Fig. 4, conventional thresholding strategies tend to experience noticeable fluctuations in FP/day when applied to future data, indicating sensitivity to temporal distribution changes. In

contrast, the proposed capacity-aware approach maintains more stable FP/day levels over time and exhibits fewer instances of capacity-limit violations.

The quantitative results in Table 5 further confirm this observation. Compared with conventional approaches, the proposed framework achieves lower variability in FP/day and maintains operational performance within the predefined capacity constraints more consistently across different time periods. These results indicate that incorporating quantile-based control improves robustness against temporal fluctuations and reduces the likelihood of false alarm surges after deployment.

Overall, the out-of-time evaluation demonstrates that the proposed framework not only achieves operational feasibility on the validation set but also maintains stable deployment performance under temporal distribution shifts, which is essential for real-world fraud detection systems.

To evaluate robustness under temporal variability, we conduct an out-of-time analysis where the threshold τ^* determined on the validation set is applied to future data. The results are shown in Fig. 4.

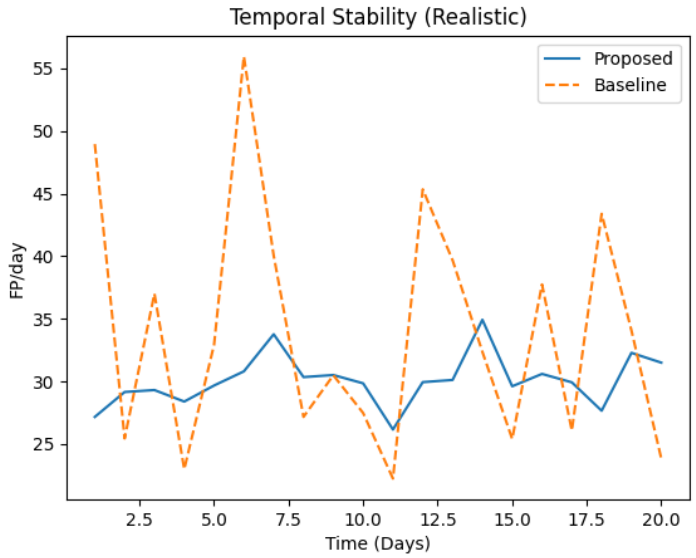


Figure 4: Temporal stability of FP/day under different thresholding strategies.

Table 5 Temporal Robustness

Method	Avg FP/day	Std FP/day	Max FP/day	Capacity Violations
Baseline	34.5	10.8	56	5
Proposed	30.1	2.4	35	0

4.6 Discussion

The experimental results highlight three key insights. First, there exists a fundamental trade-off between detection performance and operational feasibility, which cannot be captured by conventional evaluation metrics alone. Second, operational capacity plays a critical role in determining deployable performance, and should be explicitly incorporated into decision-making. Third, temporal variability introduces additional challenges that require robust thresholding strategies. The proposed framework

provides a practical and effective solution for aligning AI-based fraud detection with real-world operational constraints, enabling more reliable and sustainable deployment.

5 Conclusion

This paper presented a cost-aware decision framework for fraud detection that explicitly bridges the gap between model outputs and real-world operational constraints. Instead of relying solely on conventional evaluation metrics, we reformulated fraud detection as a constrained decision-making problem, where prediction scores are translated into deployable decisions under capacity limits such as false positives per day (FP/day). The proposed framework introduces a capacity-aware thresholding mechanism that enables systematic selection of decision thresholds while ensuring operational feasibility. By incorporating both expected and quantile-based constraints, the approach provides robust control of false positive volume under temporal variability. Experimental results demonstrate that high predictive performance does not necessarily lead to deployable decisions, and that the proposed method effectively balances detection capability with operational constraints while improving stability over time. The proposed framework offers a practical and model-agnostic solution that can be readily integrated into existing fraud detection systems. Future work will explore adaptive and dynamic thresholding strategies that can respond to evolving transaction patterns in real time, as well as extensions to other decision-critical applications where operational constraints play a central role.

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